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Інженерія програмного забезпечення

Практична робота №3

«Структурні шаблони проектування ПЗ. Composite, Decorator, Proxy, Flyweight, Adapter, Bridge, Facade»

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Залікова книжка № 4106

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Тема: «Структурні шаблони проектування ПЗ. Composite, Decorator, Proxy, Flyweight, Adapter, Bridge, Facade».

Мета: Ознайомлення з видами шаблонів проектування ПЗ. Вивчення структурних шаблонів. Отримання базових навичок з застосування шаблонів Composite, Decorator, Proxy, Flyweight, Adapter, Bridge, Facade.

Практична частина (Завдання №1)

Мій номер залікової книги: 4106, звідси $4106 \bmod 12 = 2$. Тому умова до 2 завдання буде:

Клас `GameElement`, який представляє елемент ігрового простору.

Необхідно розширити функціональність так, щоб:

- Можна було створювати простий елемент (наприклад, одиночний об'єкт на рівні).
- Кожний елемент мав розмір (ширина та висота) у умовних одиницях.
- Можна було створювати композитний елемент (групу елементів), у якому:
 - Атрибути розміру та площа обчислюються динамічно на основі вкладених елементів.
 - Можливі вкладені композитні елементи (багаторівнева ієрархія).
- Можна було отримати площу елемента:
 - Для простого елемента – площа = ширина $\times$ висота.
 - Для композитного елемента – площа = сума площ усіх вкладених елементів або мінімальна обгортка, що охоплює всі піделементи (залежно від реалізації).

Вимоги

- Не змінювати існуючий клас `GameElement`. Реалізувати можливість гнучкого додавання нових типів елементів у майбутньому.
- Продемонструвати у методі `main` кілька варіантів роботи з елементами:
 - окремий простий елемент,
 - група з кількох простих елементів,
 - група з вкладеними групами (багаторівнева структура).

Загальна структура проекту до компіляції:

```
1 PS C:\Users\Artem Davydchuk\Desktop\reps\university_roadmap\semestr3\software
2 ↪ engineering\labs\lab3\project1> tree /f /a
3 Folder PATH listing
4 Volume serial number is 9873-FF92
5 C:..
6 |   build.xml
7 |
8 \---src
9     AreaMode.java
10    CompositeElement.java
11    IGameElement.java
12    Main.java
    SimpleElement.java
```

Код build.xml задля автоматизації:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project name="lab3: First Task" default="all" basedir=".">
3
4     <!-- ===== Properties ===== -->
5     <property name="src.dir" value="src"/>
6     <property name="build.dir" value="out"/>
7     <property name="classes.dir" value="${build.dir}"/>
8     <property name="doc.dir" value="doc"/>
9     <property name="main.class" value="Main"/>
10    <property name="java.release" value="17"/>
11
12    <!-- ===== Clean ===== -->
13    <target name="clean" description="Remove compiled classes and docs">
14        <delete dir="${build.dir}"/>
15        <delete dir="${doc.dir}"/>
16    </target>
17
18    <!-- ===== Compile ===== -->
19    <target name="compile" description="Compile sources" depends="prepare">
20        <mkdir dir="${classes.dir}"/>
21        <javac srcdir="${src.dir}"
22            destdir="${classes.dir}"
23            includeantruntime="false"
24            debug="true"
25            encoding="UTF-8"
26            release="${java.release}">
27            <include name="**/*.java"/>
28        </javac>
29    </target>
30
31    <target name="prepare">
32        <mkdir dir="${classes.dir}"/>
33    </target>
34
35    <!-- ===== Run ===== -->
36    <target name="run" description="Run main class" depends="compile">
37        <java classname="${main.class}" fork="true" failonerror="true">
38            <classpath>
39                <path location="${classes.dir}"/>
40            </classpath>
41        </java>
42    </target>
43
44    <!-- ===== Javadoc ===== -->
45    <target name="javadoc" description="Generate Javadoc">
46        <mkdir dir="${doc.dir}"/>
47        <javadoc destdir="${doc.dir}"
48            sourcepath="${src.dir}"
49            encoding="UTF-8"
50            author="true"
51            version="true"
52            use="true"
53            windowtitle="Composite Pattern - Lab 3 API">
```

```

54         doctitle="Composite Pattern - Lab 3 API"
55         failonerror="false">
56         <fileset dir="${src.dir}">
57             <include name="**/*.java"/>
58         </fileset>
59         <link
60             ↪ href="https://docs.oracle.com/en/java/javase/${java.release}/docs/api/"/>
61         </javadoc>
62     </target>
63
64     <!-- ===== Convenience: all-in-one ===== -->
65     <target name="all" description="Clean, compile, run, and generate Javadoc"
66         depends="clean,compile,run,javadoc"/>
67 </project>

```

Функціонал (таргети), який реалізує build.xml

```

1 Buildfile: C:\Users\Artem
  ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
  ↪ engineering\labs\lab3\project1\build.xml
2
3 Main targets:
4
5 all      Clean, compile, run, and generate Javadoc
6 clean    Remove compiled classes and docs
7 compile  Compile sources
8 javadoc  Generate Javadoc
9 run      Run main class
10 Default target: all

```

Код AreaMode.java:

```

1 /**
2  * Area calculation modes for a composite element.
3  */
4 public enum AreaMode {
5     /** Area equals the sum of all child areas. */
6     SUM_CHILD_AREAS,
7
8     /** Area equals the area of the minimal bounding box covering all children. */
9     BOUNDING_BOX
10 }

```

Код CompositeElement.java:

```

1 import java.util.ArrayList;
2 import java.util.List;
3
4 /**
5  * Composite element (group) in the {@code Composite} pattern.
6  * <p>
7  * Holds child elements, supports nested composites, and computes size dynamically

```

```

8  * from children. Area calculation is configurable via {@link AreaMode}.
9  */
10 public class CompositeElement implements IGameElement {
11     private final String name;
12     private int x, y; // local position in the parent
13     private final List<IGameElement> children = new ArrayList<>();
14     private AreaMode areaMode = AreaMode.SUM_CHILD_AREAS;
15
16     /**
17      * Creates a composite with a human-readable name.
18      * @param name composite name
19      */
20     public CompositeElement(String name) { this.name = name; }
21
22     /**
23      * Sets the area calculation mode.
24      * @param mode mode to use
25      * @return this composite (for fluent chaining)
26      */
27     public CompositeElement setAreaMode(AreaMode mode) {
28         this.areaMode = mode;
29         return this;
30     }
31
32     /**
33      * Adds a child at the given local position within this composite.
34      * @param child element to add (simple or composite)
35      * @param childLocalX child local X
36      * @param childLocalY child local Y
37      */
38     public void add(IGameElement child, int childLocalX, int childLocalY) {
39         child.setPosition(childLocalX, childLocalY);
40         children.add(child);
41     }
42
43     /**
44      * Removes the given child if present.
45      * @param child child to remove
46      */
47     public void remove(IGameElement child) {
48         children.remove(child);
49     }
50
51     /**
52      * @return internal list of children (treat as read-only)
53      */
54     public List<IGameElement> getChildren() { return children; }
55
56     /** {@inheritDoc} */
57     @Override public String name() { return name; }
58
59     /** {@inheritDoc} */
60     @Override public int getX() { return x; }
61

```

```

62  /** {@inheritDoc} */
63  @Override public int getY() { return y; }
64
65  /** {@inheritDoc} */
66  @Override public void setPosition(int x, int y) { this.x = x; this.y = y; }
67
68  /**
69   * Computes width as the width of the minimal bounding box enclosing all
70   * ↪ children.
71   * Empty composite has width 0.
72   */
73  @Override
74  public int getWidth() {
75      if (children.isEmpty()) return 0;
76      int minX = Integer.MAX_VALUE, maxX = Integer.MIN_VALUE;
77      for (var c : children) {
78          int left = c.getX();
79          int right = c.getX() + c.getWidth();
80          minX = Math.min(minX, left);
81          maxX = Math.max(maxX, right);
82      }
83      return Math.max(0, maxX - minX);
84
85  /**
86   * Computes height as the height of the minimal bounding box enclosing all
87   * ↪ children.
88   * Empty composite has height 0.
89   */
90  @Override
91  public int getHeight() {
92      if (children.isEmpty()) return 0;
93      int minY = Integer.MAX_VALUE, maxY = Integer.MIN_VALUE;
94      for (var c : children) {
95          int top = c.getY();
96          int bottom = c.getY() + c.getHeight();
97          minY = Math.min(minY, top);
98          maxY = Math.max(maxY, bottom);
99      }
100     return Math.max(0, maxY - minY);
101
102  /**
103   * Computes area according to {@link AreaMode}.
104   * SUM_CHILD_AREAS: sum of child areas.
105   * BOUNDING_BOX: getWidth() * getHeight().
106   */
107  @Override
108  public int area() {
109      if (areaMode == AreaMode.SUM_CHILD_AREAS) {
110          int sum = 0;
111          for (var c : children) sum += c.area();
112          return sum;
113      } else {

```

```

114         return getWidth() * getHeight();
115     }
116 }
117
118 /** {@inheritDoc} */
119 @Override
120 public void draw(int x, int y) {
121     int absX = x + this.x;
122     int absY = y + this.y;
123     System.out.printf(
124         "Composite.draw name=%s at=(%d,%d) size=%dx%d mode=%s children=%d%n",
125         name, absX, absY, getWidth(), getHeight(), areaMode, children.size()
126     );
127     for (var c : children) c.draw(absX, absY);
128 }
129 }

```

Код IGameElement.java:

```

1  /**
2   * Base interface for an element of the game space.
3   * <p>
4   * Serves as the <b>Component</b> role in the <em>Composite</em> pattern,
5   * defining a common contract for both simple and composite elements.
6   */
7  public interface IGameElement {
8
9      /** @return human-readable name (for logs/debugging) */
10     String name();
11
12     /** @return local X coordinate within parent composite */
13     int getX();
14
15     /** @return local Y coordinate within parent composite */
16     int getY();
17
18     /**
19      * Sets the local position of this element relative to its parent composite.
20      * @param x local X
21      * @param y local Y
22      */
23     void setPosition(int x, int y);
24
25     /**
26      * Width of the element.
27      * For composites, this is computed dynamically from the children's bounding
28      * ↪ box.
29      * @return non-negative width
30      */
31     int getWidth();
32
33     /**
34      * Height of the element.

```

```

34     * For composites, this is computed dynamically from the children's bounding
    ↪ box.
35     * @return non-negative height
36     */
37     int getHeight();
38
39     /**
40     * Area of the element.
41     * For composites, the calculation depends on {@link AreaMode}.
42     * @return non-negative area
43     */
44     int area();
45
46     /**
47     * Demonstrational business method (drawing/logging).
48     * Should print an informative line to the console.
49     * @param x absolute X coordinate of the parent origin
50     * @param y absolute Y coordinate of the parent origin
51     */
52     void draw(int x, int y);
53 }

```

Код SimpleElement.java:

```

1  /**
2   * Leaf element in the {@code Composite} pattern.
3   * <p>
4   * Represents an indivisible game object with fixed size
5   * (width × height) and a local position within its parent composite.
6   */
7  public class SimpleElement implements IGameElement {
8      private final String name;
9      private int x, y;           // local position in parent
10     private final int width;
11     private final int height;
12
13     /**
14     * Creates a simple element with given name and size.
15     * Negative width/height are clamped to zero.
16     *
17     * @param name    element name (for logs/debugging)
18     * @param width    width (non-negative)
19     * @param height   height (non-negative)
20     */
21     public SimpleElement(String name, int width, int height) {
22         this.name = name;
23         this.width = Math.max(0, width);
24         this.height = Math.max(0, height);
25     }
26
27     /** {@inheritDoc} */
28     @Override public String name() { return name; }
29
30     /** {@inheritDoc} */

```

```

31     @Override public int getX() { return x; }
32
33     /** {@inheritDoc} */
34     @Override public int getY() { return y; }
35
36     /** {@inheritDoc} */
37     @Override public void setPosition(int x, int y) { this.x = x; this.y = y; }
38
39     /** {@inheritDoc} */
40     @Override public int getWidth() { return width; }
41
42     /** {@inheritDoc} */
43     @Override public int getHeight() { return height; }
44
45     /** {@inheritDoc} */
46     @Override public int area() { return width * height; }
47
48     /** {@inheritDoc} */
49     @Override
50     public void draw(int x, int y) {
51         System.out.printf("Simple.draw name=%s at=(%d,%d) size=%dx%d%n",
52             name, x + this.x, y + this.y, width, height);
53     }
54 }

```

Код Main.java:

```

1  /**
2   * Demo/Client that showcases the Composite pattern implementation
3   * with three scenarios required by the lab:
4   * <ol>
5   *   <li>single simple element;</li>
6   *   <li>composite with several simple elements;</li>
7   *   <li>nested composite structure (composite inside composite)
8   *     and different area modes.</li>
9   * </ol>
10  */
11  public class Main {
12      /**
13       * Program entry point.
14       * @param args unused
15       */
16      public static void main(String[] args) {
17          // 1) Single simple element
18          IGameElement hero = new SimpleElement("Hero", 2, 3);
19          hero.setPosition(5, 7);
20          hero.draw(0, 0);
21          System.out.printf("Hero area=%d size=%dx%d%n%n", hero.area(),
22              ↪ hero.getWidth(), hero.getHeight());
23
24          // 2) Composite with several simple elements (area mode: SUM_CHILD_AREAS)
25          CompositeElement group1 = new
26              ↪ CompositeElement("Group1").setAreaMode(AreaMode.SUM_CHILD_AREAS);
27          group1.add(new SimpleElement("Tree", 3, 4), 0, 0);

```

```

26     group1.add(new SimpleElement("Rock", 2, 2), 4, 1);
27     group1.add(new SimpleElement("Chest", 1, 1), 7, 0);
28     group1.setPosition(10, 10);
29     group1.draw(0, 0);
30     System.out.printf("Group1 area=%d size=%dx%d%n%n", group1.area(),
        ↪ group1.getWidth(), group1.getHeight());
31
32     // 3) Nested composite (outer uses BOUNDING_BOX, inner uses SUM_CHILD_AREAS)
33     CompositeElement inner = new
        ↪ CompositeElement("Inner").setAreaMode(AreaMode.SUM_CHILD_AREAS);
34     inner.add(new SimpleElement("Coin", 1, 1), 1, 0);
35     inner.add(new SimpleElement("Key", 1, 2), 2, 2);
36
37     CompositeElement outer = new
        ↪ CompositeElement("Outer").setAreaMode(AreaMode.BOUNDING_BOX);
38     outer.add(new SimpleElement("Door", 2, 5), 0, 0);
39     outer.add(inner, 5, 1);
40     outer.setPosition(20, 3);
41
42     outer.draw(0, 0);
43     System.out.printf("Outer area(mode=%s)=%d size=%dx%d%n",
44         AreaMode.BOUNDING_BOX, outer.area(), outer.getWidth(),
        ↪ outer.getHeight());
45 }
46 }

```

Результат виконання команди all:

```

1 PS C:\Users\Artem Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1> ant all
2 Buildfile: C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\build.xml
3
4 clean:
5
6 prepare:
7     [mkdir] Created dir: C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\out
8
9 compile:
10    [javac] Compiling 5 source files to C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\out
11
12 run:
13    [java] Simple.draw name=Hero at=(5,7) size=2x3
14    [java] Hero area=6 size=2x3
15    [java]
16    [java] Composite.draw name=Group1 at=(10,10) size=8x4 mode=SUM_CHILD_AREAS
   ↪ children=3
17    [java] Simple.draw name=Tree at=(10,10) size=3x4
18    [java] Simple.draw name=Rock at=(14,11) size=2x2

```

```

19     [java] Simple.draw name=Chest at=(17,10) size=1x1
20     [java] Group1 area=17 size=8x4
21     [java]
22     [java] Composite.draw name=Outer at=(20,3) size=7x5 mode=BOUNDING_BOX
23         ↪ children=2
24     [java] Simple.draw name=Door at=(20,3) size=2x5
25     [java] Composite.draw name=Inner at=(25,4) size=2x4 mode=SUM_CHILD_AREAS
26         ↪ children=2
27     [java] Simple.draw name=Coin at=(26,4) size=1x1
28     [java] Simple.draw name=Key at=(27,6) size=1x2
29     [java] Outer area(mode=BOUNDING_BOX)=35 size=7x5
30
31 javadoc:
32     [mkdir] Created dir: C:\Users\Artem
33         ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
34         ↪ engineering\labs\lab3\project1\doc
35 [javadoc] C:\Users\Artem
36     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
37     ↪ engineering\labs\lab3\project1\src contains source files in the default
38     ↪ package, you must specify them as source files not packages.
39 [javadoc] Generating Javadoc
40 [javadoc] Javadoc execution
41 [javadoc] Loading source file C:\Users\Artem
42     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
43     ↪ engineering\labs\lab3\project1\src\AreaMode.java...
44 [javadoc] Loading source file C:\Users\Artem
45     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
46     ↪ engineering\labs\lab3\project1\src\CompositeElement.java...
47 [javadoc] Loading source file C:\Users\Artem
48     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
49     ↪ engineering\labs\lab3\project1\src\IGameElement.java...
50 [javadoc] Loading source file C:\Users\Artem
51     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
52     ↪ engineering\labs\lab3\project1\src\Main.java...
53 [javadoc] Loading source file C:\Users\Artem
54     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
55     ↪ engineering\labs\lab3\project1\src\SimpleElement.java...
56 [javadoc] Constructing Javadoc information...
57 [javadoc] Building index for all the packages and classes...
58 [javadoc] Standard Doclet version 21.0.8+12-LTS-250
59 [javadoc] Building tree for all the packages and classes...
60 [javadoc] Generating C:\Users\Artem
61     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
62     ↪ engineering\labs\lab3\project1\doc\AreaMode.html...
63 [javadoc] Generating C:\Users\Artem
64     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
65     ↪ engineering\labs\lab3\project1\doc\CompositeElement.html...
66 [javadoc] C:\Users\Artem
67     ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
68     ↪ engineering\labs\lab3\project1\src\IGameElement.java:9: warning: no main
69     ↪ description
70 [javadoc]      /** @return human-readable name (for logs/debugging) */
71 [javadoc]      ~

```

```

48 [javadoc] C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\src\IGameElement.java:12: warning: no main
   ↪ description
49 [javadoc]      /** @return local X coordinate within parent composite */
50 [javadoc]      ~
51 [javadoc] C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\src\IGameElement.java:15: warning: no main
   ↪ description
52 [javadoc]      /** @return local Y coordinate within parent composite */
53 [javadoc]      ~
54 [javadoc] C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\src\CompositeElement.java:52: warning: no main
   ↪ description
55 [javadoc]      * @return internal list of children (treat as read-only)
56 [javadoc]      ~
57 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\IGameElement.html...
58 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\Main.html...
59 [javadoc] C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\src\Main.java:11: warning: use of default
   ↪ constructor, which does not provide a comment
60 [javadoc] public class Main {
61 [javadoc]      ~
62 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\SimpleElement.html...
63 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\package-summary.html...
64 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\package-tree.html...
65 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\class-use\AreaMode.html...
66 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\class-use\CompositeElement.html...
67 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\class-use\IGameElement.html...
68 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\class-use\Main.html...
69 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\class-use\SimpleElement.html...

```

```

70 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\package-use.html...
71 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\overview-tree.html...
72 [javadoc] Building index for all classes...
73 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\allclasses-index.html...
74 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\allpackages-index.html...
75 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\index-all.html...
76 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\search.html...
77 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\index.html...
78 [javadoc] Generating C:\Users\Artem
   ↪ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1\doc\help-doc.html...
79 [javadoc] 5 warnings
80
81 all:
82
83 BUILD SUCCESSFUL
84 Total time: 5 seconds

```

Оновлене дерево робочої директорії:

```

1 PS C:\Users\Artem Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1> tree /f /a
2 Folder PATH listing
3 Volume serial number is 9873-FF92
4 C:.
5 |   build.xml
6 |
7 +---doc
8 |   |   allclasses-index.html
9 |   |   allpackages-index.html
10 |   |   AreaMode.html
11 |   |   CompositeElement.html
12 |   |   copy.svg
13 |   |   element-list
14 |   |   help-doc.html
15 |   |   IGameElement.html
16 |   |   index-all.html
17 |   |   index.html
18 |   |   link.svg
19 |   |   Main.html

```

```

20 | | member-search-index.js
21 | | module-search-index.js
22 | | overview-tree.html
23 | | package-search-index.js
24 | | package-summary.html
25 | | package-tree.html
26 | | package-use.html
27 | | script.js
28 | | search-page.js
29 | | search.html
30 | | search.js
31 | | SimpleElement.html
32 | | stylesheet.css
33 | | tag-search-index.js
34 | | type-search-index.js
35 | |
36 | +---class-use
37 | |     AreaMode.html
38 | |     CompositeElement.html
39 | |     IGameElement.html
40 | |     Main.html
41 | |     SimpleElement.html
42 | |
43 | +---legal
44 | |     COPYRIGHT
45 | |     jquery.md
46 | |     jqueryUI.md
47 | |     LICENSE
48 | |
49 | +---resources
50 | |     glass.png
51 | |     x.png
52 | |
53 | \---script-dir
54 |     jquery-3.7.1.min.js
55 |     jquery-ui.min.css
56 |     jquery-ui.min.js
57 |
58 +---out
59 |     AreaMode.class
60 |     CompositeElement.class
61 |     IGameElement.class
62 |     Main.class
63 |     SimpleElement.class
64 |
65 \---src
66     AreaMode.java
67     CompositeElement.java
68     IGameElement.java
69     Main.java
70     SimpleElement.java

```

Додатковий окремий запуск Main:

```
1 PS C:\Users\Artem Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project1> java -cp out Main
2 Simple.draw name=Hero at=(5,7) size=2x3
3 Hero area=6 size=2x3
4
5 Composite.draw name=Group1 at=(10,10) size=8x4 mode=SUM_CHILD_AREAS children=3
6 Simple.draw name=Tree at=(10,10) size=3x4
7 Simple.draw name=Rock at=(14,11) size=2x2
8 Simple.draw name=Chest at=(17,10) size=1x1
9 Group1 area=17 size=8x4
10
11 Composite.draw name=Outer at=(20,3) size=7x5 mode=BOUNDING_BOX children=2
12 Simple.draw name=Door at=(20,3) size=2x5
13 Composite.draw name=Inner at=(25,4) size=2x4 mode=SUM_CHILD_AREAS children=2
14 Simple.draw name=Coin at=(26,4) size=1x1
15 Simple.draw name=Key at=(27,6) size=1x2
16 Outer area(mode=BOUNDING_BOX)=35 size=7x5
```

Відкриваємо index.html у браузері:

PACKAGE

CLASS

USE

TREE

INDEX

HELP

PACKAGE: DESCRIPTION | RELATED PACKAGES | CLASSES AND INTERFACES

Unnamed Package

All Classes and Interfaces

Interfaces

Classes

Enum Classes

Class	Description
AreaMode	Area calculation modes for a composite element.
CompositeElement	Composite element (group) in the Composite pattern.
IGameElement	Base interface for an element of the game space.
Main	Demo/Client that showcases the Composite pattern implementation (simple composite) and different area modes.
SimpleElement	Leaf element in the Composite pattern.

Рис. 1

Практична частина (Завдання №2)

Номер залікової: 4106, $4106 \bmod 11 = 3$. Звідси завдання:

Визначити специфікації класів, які подають об'єкти-іконки для зображення елементів файлової системи при побудові графічного інтерфейсу користувача (GUI) – примітиви (файли) та їх композиції (директорії). Забезпечити ефективне використання пам'яті при роботі з великою кількістю графічних об'єктів. Реалізувати метод рисування графічного об'єкту.

У цьому завданні використаю структурний шаблон проектування Flyweight (Легковаговик), який забезпечує ефективне використання пам'яті при роботі з великою кількістю графічних об'єктів (іконок файлів і директорій). Також частково застосовано Composite (Компонувальник) для побудови ієрархії елементів файлової системи.

Дерево робочої директорії до виконання build:

```
1 PS C:\Users\Artem Davydchuk\Desktop\reps\university_roadmap\semestr3\software
   ↪ engineering\labs\lab3\project2> tree /f /a
2 Folder PATH listing
3 Volume serial number is 9873-FF92
4 C:.
5 |   build.xml
6 |
7 \---src
8     DirectoryIcon.java
9     FileIcon.java
10    FSIcon.java
11    IconFactory.java
12    IconFlyweight.java
13    IconType.java
14    Main.jav
```

Код build.xml задля автоматизації:

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project name="lab3: Second Task" default="all" basedir=".">
3
4   <!-- ===== Properties ===== -->
5   <property name="src.dir"      value="src"/>
6   <property name="build.dir"    value="out"/>
7   <property name="classes.dir"  value="${build.dir}"/>
8   <property name="doc.dir"      value="doc"/>
9   <property name="main.class"   value="Main"/>
10  <property name="java.release" value="17"/>
11
12  <!-- ===== Clean ===== -->
13  <target name="clean" description="Remove compiled classes and docs">
14    <delete dir="${build.dir}"/>
15    <delete dir="${doc.dir}"/>
16  </target>
17
```

```

18 <!-- ===== Compile ===== -->
19 <target name="compile" description="Compile sources" depends="prepare">
20     <mkdir dir="${classes.dir}"/>
21     <javac srcdir="${src.dir}"
22         destdir="${classes.dir}"
23         includeantruntime="false"
24         debug="true"
25         encoding="UTF-8"
26         release="${java.release}">
27         <include name="**/*.java"/>
28         <!-- Якщо з'явиться зовнішній classpath (JAR-и) - додай тут <classpath> -->
29     </javac>
30 </target>
31
32 <target name="prepare">
33     <mkdir dir="${classes.dir}"/>
34 </target>
35
36 <!-- ===== Run ===== -->
37 <target name="run" description="Run main class" depends="compile">
38     <java classname="${main.class}" fork="true" failonerror="true">
39         <classpath>
40             <path location="${classes.dir}"/>
41         </classpath>
42     </java>
43 </target>
44
45 <!-- ===== Javadoc ===== -->
46 <target name="javadoc" description="Generate Javadoc">
47     <mkdir dir="${doc.dir}"/>
48     <javadoc destdir="${doc.dir}"
49         sourcepath="${src.dir}"
50         encoding="UTF-8"
51         author="true"
52         version="true"
53         use="true"
54         windowtitle="IPZ - API documentation"
55         doctitle="IPZ - API documentation"
56         failonerror="false">
57         <fileset dir="${src.dir}">
58             <include name="**/*.java"/>
59         </fileset>
60         <link
61             ↪ href="https://docs.oracle.com/en/java/javase/${java.release}/docs/api/">
62     </javadoc>
63 </target>
64
65 <!-- ===== All-in-one ===== -->
66 <target name="all" description="Clean, compile, run, javadoc"
67     depends="clean,compile,run,javadoc"/>
68 </project>

```

Команди (таски) які він виконує:

```
1 Buildfile: C:\Users\Artem
  ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
  ↳ engineering\labs\lab3\project2\build.xml
2
3 Main targets:
4
5 all      Clean, compile, run, javadoc
6 clean    Remove compiled classes and docs
7 compile  Compile sources
8 javadoc  Generate Javadoc
9 run      Run main class
10 Default target: all
```

Результат виконання команди all:

```
1 PS C:\Users\Artem Davydchuk\Desktop\reps\university_roadmap\semestr3\software
  ↳ engineering\labs\lab3\project2> ant all
2 Buildfile: C:\Users\Artem
  ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
  ↳ engineering\labs\lab3\project2\build.xml
3
4 clean:
5
6 prepare:
7   [mkdir] Created dir: C:\Users\Artem
  ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
  ↳ engineering\labs\lab3\project2\out
8
9 compile:
10  [javac] Compiling 7 source files to C:\Users\Artem
  ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
  ↳ engineering\labs\lab3\project2\out
11
12 run:
13  [java] draw[FOLDER sprite=sprite-folder size=28x24] at=(10,10) label="root/"
14  [java] draw[FILE sprite=sprite-file size=24x24] at=(10,40) label="README.md"
15  [java] draw[FILE sprite=sprite-file size=24x24] at=(44,40) label=".gitignore"
16  [java] draw[FOLDER sprite=sprite-folder size=28x24] at=(78,40) label="src/"
17  [java] draw[FILE sprite=sprite-file size=24x24] at=(78,70) label="FSIcon.java"
18  [java] draw[FILE sprite=sprite-file size=24x24] at=(112,70)
  ↳ label="IconFlyweight.java"
19  [java] draw[FILE sprite=sprite-file size=24x24] at=(146,70)
  ↳ label="IconFactory.java"
20  [java] draw[FILE sprite=sprite-file size=24x24] at=(180,70)
  ↳ label="FileIcon.java"
21  [java] draw[FILE sprite=sprite-file size=24x24] at=(214,70)
  ↳ label="DirectoryIcon.java"
22  [java] draw[FILE sprite=sprite-file size=24x24] at=(248,70) label="Main.java"
23  [java] draw[FOLDER sprite=sprite-folder size=28x24] at=(112,40) label="doc/"
24  [java] draw[FILE sprite=sprite-file size=24x24] at=(112,70) label="report.tex"
25  [java] draw[FILE sprite=sprite-file size=24x24] at=(146,70) label="build.xml"
```

```

26     [java] draw[FILE sprite=sprite-file size=24x24] at=(180,70)
      ↪ label="diagram.puml"
27     [java]
28     [java] Children in /src: 5006
29     [java] Flyweights cached: 2 (should be 2: FILE and FOLDER)
30     [java]
31     [java] [Preview: redraw /src header only]
32     [java] draw[FOLDER sprite=sprite-folder size=28x24] at=(10,200) label="src/"
33     [java] ... (thousands of file icons share the same FILE flyweight)
34
35 javadoc:
36     [mkdir] Created dir: C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\doc
37 [javadoc] C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src contains source files in the default
      ↪ package, you must specify them as source files not packages.
38 [javadoc] Generating Javadoc
39 [javadoc] Javadoc execution
40 [javadoc] Loading source file C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src\DirectoryIcon.java...
41 [javadoc] Loading source file C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src\FIcon.java...
42 [javadoc] Loading source file C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src\FileIcon.java...
43 [javadoc] Loading source file C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src\IconFactory.java...
44 [javadoc] Loading source file C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src\IconFlyweight.java...
45 [javadoc] Loading source file C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src\IconType.java...
46 [javadoc] Loading source file C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src\Main.java...
47 [javadoc] Constructing Javadoc information...
48 [javadoc] Building index for all the packages and classes...
49 [javadoc] Standard Doclet version 21.0.8+12-LTS-250
50 [javadoc] Building tree for all the packages and classes...
51 [javadoc] C:\Users\Artem
      ↪ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
      ↪ engineering\labs\lab3\project2\src\FileIcon.java:4: error: bad HTML entity
52 [javadoc] * Extrinsic state: file name, local position; intrinsic: shared sprite
      ↪ & size.
53 [javadoc]
      ↪ ^

```

```

54 [javadoc] Generating C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\doc\DirectoryIcon.html...
55 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\FIcon.java:7: warning: no main description
56 [javadoc]      /** @return display name (file or directory name) */
57 [javadoc]      ~
58 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\FIcon.java:11: warning: no @return
59 [javadoc]      int getX();
60 [javadoc]      ~
61 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\FIcon.java:16: warning: no @return
62 [javadoc]      int getWidth();
63 [javadoc]      ~
64 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\DirectoryIcon.java:27: warning: no @param
   ↳ for child
65 [javadoc]      public void add(FIcon child) {
66 [javadoc]      ~
67 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\DirectoryIcon.java:41: warning: no @param
   ↳ for child
68 [javadoc]      public void remove(FIcon child) { children.remove(child); }
69 [javadoc]      ~
70 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\DirectoryIcon.java:43: warning: no main
   ↳ description
71 [javadoc]      /** @return immutable view of children */
72 [javadoc]      ~
73 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\DirectoryIcon.java:47: warning: no @param
   ↳ for columns
74 [javadoc]      public DirectoryIcon setGrid(int columns, int hGap, int vGap) {
75 [javadoc]      ~
76 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\DirectoryIcon.java:47: warning: no @param
   ↳ for hGap
77 [javadoc]      public DirectoryIcon setGrid(int columns, int hGap, int vGap) {
78 [javadoc]      ~
79 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
   ↳ engineering\labs\lab3\project2\src\DirectoryIcon.java:47: warning: no @param
   ↳ for vGap
80 [javadoc]      public DirectoryIcon setGrid(int columns, int hGap, int vGap) {
81 [javadoc]      ~

```

```

82 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\src\DirectoryIcon.java:47: warning: no @return
83 [javadoc]     public DirectoryIcon setGrid(int columns, int hGap, int vGap) {
84 [javadoc]         ~
85 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\src\DirectoryIcon.java:21: warning: no comment
86 [javadoc]     public DirectoryIcon(String dirName) {
87 [javadoc]         ~
88 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\src\FSIcon.java:12: warning: no comment
89 [javadoc]     int getY();
90 [javadoc]         ~
91 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\src\FSIcon.java:13: warning: no comment
92 [javadoc]     void setPosition(int x, int y);
93 [javadoc]         ~
94 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\src\FSIcon.java:17: warning: no comment
95 [javadoc]     int getHeight();
96 [javadoc]         ~
97 [javadoc] Generating C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\doc\FileIcon.html...
98 [javadoc] Generating C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\doc\FSIcon.html...
99 [javadoc] Generating C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\doc\IconFactory.html...
100 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\src\IconFactory.java:14: warning: no @param for
   ↳ type
101 [javadoc]     public static IconFlyweight get(IconType type) {
102 [javadoc]         ~
103 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\src\IconFactory.java:14: warning: no @return
104 [javadoc]     public static IconFlyweight get(IconType type) {
105 [javadoc]         ~
106 [javadoc] C:\Users\Artem
   ↳ Davydchuk\Desktop\repos\university_roadmap\semeestr3\software
   ↳ engineering\labs\lab3\project2\src\IconFactory.java:27: warning: no main
   ↳ description
107 [javadoc]     /** @return how many flyweights are currently cached (should be
   ↳ small). */
108 [javadoc]         ~

```

```

109 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\IconFlyweight.html...
110 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconFlyweight.java:18: warning: no main
    ↳ description
111 [javadoc]      /** @return icon type (FILE/FOLDER) */
112 [javadoc]      ~
113 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconFlyweight.java:21: warning: no main
    ↳ description
114 [javadoc]      /** @return logical sprite identifier (e.g., resource name) */
115 [javadoc]      ~
116 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconFlyweight.java:24: warning: no main
    ↳ description
117 [javadoc]      /** @return intrinsic width */
118 [javadoc]      ~
119 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconFlyweight.java:27: warning: no main
    ↳ description
120 [javadoc]      /** @return intrinsic height */
121 [javadoc]      ~
122 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconFlyweight.java:34: warning: no @param
    ↳ for label
123 [javadoc]      public void paint(String label, int absX, int absY) {
124 [javadoc]      ~
125 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconFlyweight.java:34: warning: no @param
    ↳ for absX
126 [javadoc]      public void paint(String label, int absX, int absY) {
127 [javadoc]      ~
128 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconFlyweight.java:34: warning: no @param
    ↳ for absY
129 [javadoc]      public void paint(String label, int absX, int absY) {
130 [javadoc]      ~
131 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconFlyweight.java:11: warning: no comment
132 [javadoc]      public IconFlyweight(IconType type, String spriteName, int width,
    ↳ int height) {
133 [javadoc]      ~
134 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\IconType.html...

```

```

135 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconType.java:3: warning: no comment
136 [javadoc] FILE,
137 [javadoc] ~
138 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\IconType.java:4: warning: no comment
139 [javadoc] FOLDER
140 [javadoc] ~
141 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\Main.html...
142 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\Main.java:5: warning: use of default
    ↳ constructor, which does not provide a comment
143 [javadoc] public class Main {
144 [javadoc] ~
145 [javadoc] C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\src\Main.java:6: warning: no comment
146 [javadoc] public static void main(String[] args) {
147 [javadoc] ~
148 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\package-summary.html...
149 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\package-tree.html...
150 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\class-use\DirectoryIcon.html...
151 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\class-use\FSIcon.html...
152 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\class-use\FileIcon.html...
153 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\class-use\IconFactory.html...
154 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\class-use\IconFlyweight.html...
155 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\class-use\IconType.html...
156 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\class-use\Main.html...
157 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\repos\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\package-use.html...

```

```

158 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\overview-tree.html...
159 [javadoc] Building index for all classes...
160 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\allclasses-index.html...
161 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\allpackages-index.html...
162 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\index-all.html...
163 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\search.html...
164 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\index.html...
165 [javadoc] Generating C:\Users\Artem
    ↳ Davydchuk\Desktop\reps\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2\doc\help-doc.html...
166 [javadoc] 1 error
167 [javadoc] 29 warnings
168
169 all:
170
171 BUILD SUCCESSFUL
172 Total time: 5 seconds

```

Оновлене дерево робочої директорії:

```

1 PS C:\Users\Artem Davydchuk\Desktop\reps\university_roadmap\semestr3\software
    ↳ engineering\labs\lab3\project2> tree /f /a
2 Folder PATH listing
3 Volume serial number is 9873-FF92
4 C:.
5 |   build.xml
6 |
7 +---doc
8 |   |   allclasses-index.html
9 |   |   allpackages-index.html
10 |   |   copy.svg
11 |   |   DirectoryIcon.html
12 |   |   element-list
13 |   |   FileIcon.html
14 |   |   FSIcon.html
15 |   |   help-doc.html
16 |   |   IconFactory.html
17 |   |   IconFlyweight.html
18 |   |   IconType.html
19 |   |   index-all.html
20 |   |   index.html
21 |   |   link.svg

```

```

22 | | Main.html
23 | | member-search-index.js
24 | | module-search-index.js
25 | | overview-tree.html
26 | | package-search-index.js
27 | | package-summary.html
28 | | package-tree.html
29 | | package-use.html
30 | | script.js
31 | | search-page.js
32 | | search.html
33 | | search.js
34 | | stylesheet.css
35 | | tag-search-index.js
36 | | type-search-index.js
37 | |
38 | +---class-use
39 | |     DirectoryIcon.html
40 | |     FileIcon.html
41 | |     FSIcon.html
42 | |     IconFactory.html
43 | |     IconFlyweight.html
44 | |     IconType.html
45 | |     Main.html
46 | |
47 | +---legal
48 | |     COPYRIGHT
49 | |     jquery.md
50 | |     jqueryUI.md
51 | |     LICENSE
52 | |
53 | +---resources
54 | |     glass.png
55 | |     x.png
56 | |
57 | \---script-dir
58 |     jquery-3.7.1.min.js
59 |     jquery-ui.min.css
60 |     jquery-ui.min.js
61 |
62 +---out
63 |     DirectoryIcon.class
64 |     FileIcon.class
65 |     FSIcon.class
66 |     IconFactory$1.class
67 |     IconFactory.class
68 |     IconFlyweight.class
69 |     IconType.class
70 |     Main.class
71 |
72 \---src
73     DirectoryIcon.java
74     FileIcon.java
75     FSIcon.java

```

```

76 IconFactory.java
77 IconFlyweight.java
78 IconType.java
79 Main.java

```

Додатковий окремий запуск Main:

```

1 PS C:\Users\Artem Davydchuk\Desktop\reps\university_roadmap\semestr3\software
  ↪ engineering\labs\lab3\project2> java -cp out Main
2 draw[FOLDER sprite=sprite-folder size=28x24] at=(10,10) label="root/"
3 draw[FILE sprite=sprite-file size=24x24] at=(10,40) label="README.md"
4 draw[FILE sprite=sprite-file size=24x24] at=(44,40) label=".gitignore"
5 draw[FOLDER sprite=sprite-folder size=28x24] at=(78,40) label="src/"
6 draw[FILE sprite=sprite-file size=24x24] at=(78,70) label="FSIcon.java"
7 draw[FILE sprite=sprite-file size=24x24] at=(112,70) label="IconFlyweight.java"
8 draw[FILE sprite=sprite-file size=24x24] at=(146,70) label="IconFactory.java"
9 draw[FILE sprite=sprite-file size=24x24] at=(180,70) label="FileIcon.java"
10 draw[FILE sprite=sprite-file size=24x24] at=(214,70) label="DirectoryIcon.java"
11 draw[FILE sprite=sprite-file size=24x24] at=(248,70) label="Main.java"
12 draw[FOLDER sprite=sprite-folder size=28x24] at=(112,40) label="doc/"
13 draw[FILE sprite=sprite-file size=24x24] at=(112,70) label="report.tex"
14 draw[FILE sprite=sprite-file size=24x24] at=(146,70) label="build.xml"
15 draw[FILE sprite=sprite-file size=24x24] at=(180,70) label="diagram.puml"
16
17 Children in /src: 5006
18 Flyweights cached: 2 (should be 2: FILE and FOLDER)
19
20 [Preview: redraw /src header only]
21 draw[FOLDER sprite=sprite-folder size=28x24] at=(10,200) label="src/"
22 ... (thousands of file icons share the same FILE flyweight)

```

Відкриваємо index.html у браузері:

PACKAGE CLASS USE TREE INDEX HELP	
PACKAGE: DESCRIPTION RELATED PACKAGES CLASSES AND INTERFACES	
Unnamed Package	
All Classes and Interfaces	Interfaces
Classes	Enum Classes
Class	Description
DirectoryIcon	Composite: a directory icon that contains child icons (files or directories).
FileIcon	Leaf of the Composite: represents a file icon.
FSIcon	Component for file-system icons in a GUI.
IconFactory	Flyweight factory that reuses a small set of heavy icon objects.
IconFlyweight	Flyweight that stores heavy, intrinsic icon data (sprite, size, etc.).
IconType	Types of icons supported by the flyweight cache.
Main	Demo: filesystem icons with Composite (directories) + Flyweight (shared sprites).

Рис. 2

Висновок: У результаті виконання лабораторної роботи було розроблено програмну модель, яка відображає елементи файлової системи у вигляді графічних іконок. Під час реалізації було використано структурні шаблони проектування **Flyweight** (Легкого-вик) та **Composite** (Компонувальник).

Застосування шаблону **Flyweight** забезпечило ефективне використання пам'яті завдяки спільному зберіганню важких графічних ресурсів (іконок), які використовуються численними об'єктами файлової системи. Кожен об'єкт має власний зовнішній стан (ім'я, координати розташування), але спільно використовує внутрішній стан — графічне зображення, розміри та тип піктограми.

Шаблон **Composite** дозволив реалізувати ієрархічну структуру файлової системи, де директорії можуть містити як окремі файли, так і інші вкладені директорії. Це забезпечило уніфіковану роботу з усіма об'єктами через спільний інтерфейс та спростило виконання операцій відображення.

Таким чином, поставлене завдання виконано: створено систему класів, яка забезпечує ефективну обробку та відображення великої кількості графічних об'єктів із мінімальними витратами пам'яті.